

Homework for Chapter 7

- Study with the data from any source (reference books, websites, AI apps), and compare the transverse and longitudinal eigen-frequencies of optical phonons of a series of semiconductors of ZnO, ZnS, ZnSe, and ZnTe, what do you conclude?
- Compare the longitudinal-transverse splitting at $k = 0$ for Ge and the corresponding III–V, II–VI and I–VII compounds on the same row of the periodic table of the elements. What do you find and qualitatively explain your findings.

附录5 元素周期																		0	电子层	0 族 电子数
周期	I A	II A	III A	IV A	V A	VI A	VII A	0												
1	1 H 1.008																	2 He 4.003	K	2
2	3 Li 6.941	4 Be 9.012																		
3	11 Na 22.99	12 Mg 24.31																		
4	19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.41	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80	N M L K	8 18 8 2
5	37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc [98]	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3	O N M L K	8 18 18 8 2
6	55 Cs 132.9	56 Ba 137.3	57-71 La-Lu 镧系	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.5	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po [209]	85 At [210]	86 Rn [222]	P O N M L K	8 18 32 18 8 2
7	87 Fr [223]	88 Ra [226]	89-103 Ac-Lr 锕系	104 Rf [261]	105 Db [262]	106 Sg [266]	107 Bh [264]	108 Hs [277]	109 Mt [268]	110 Uun [281]	111 Uuh [285]	112 Uub [285]	注：相对原子质量取自2001年国际原子量，并全部取4位有效数字。							
镧系	57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm [145]	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0					
锕系	89 Ac [227]	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np [237]	94 Pu [244]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]					

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	TO (THz)	LO (THz)	Δ (THz)	$\frac{\omega_{LO}}{\omega_{TO}}$
1. ZnO	11.39	17.54	6.15	1.54
ZnS	8.18	10.49	2.31	1.28
ZnSe	6.18	7.55	1.37	1.22
ZnTe	4.20	5.10	0.90	1.21

① 原子变重, 约化质量上升 (振动质量上升) TO, LO 下降

② 原子变重, $\Delta = LO - TO$ 下降, 劈裂程度取决于有效电荷数
 $O^{2-} \rightarrow Te^{2-}$ 从离子到共价晶体过渡, 电荷减小, Δ 下降

$\frac{\omega_{LO}}{\omega_{TO}} = \sqrt{\frac{\epsilon_{\infty}}{\epsilon_{\infty} - \epsilon_{\infty}}}$ 反映晶体极性强弱, 共价成分越大, 极性弱

	TO (cm ⁻¹)	LO (cm ⁻¹)	Δ (cm ⁻¹)
2. Ge	301	301	0
GaAs	268	292	24
ZnSe	206	250	44
CuBr	150	220	70

$$\Delta_{Ge} < \Delta_{GaAs} < \Delta_{ZnSe} < \Delta_{CuBr}$$

对于 LO 模, 其引起的净极化 $\left(\frac{\omega_{LO}}{\omega_{TO}}\right)^2 = \frac{\epsilon_0}{\epsilon_{\infty}}$

LO-TO 越大, 净电荷转移的发生程度更大, 即离子性更强

Ge 为纯共价晶体, 几乎不存在净极化

由 GaAs $\rightarrow$ CuBr 离子性不断增强, Δ 不断变大